

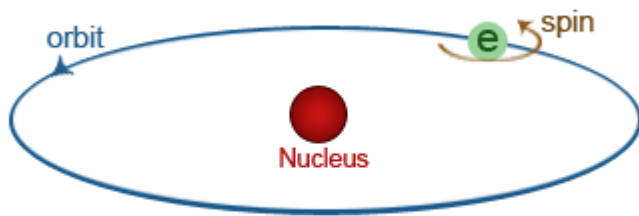
Magnetism

Magnetic force

If the charged particles (e.g. electrons) are at rest, then they will produce only electric force on the second charge. But if the charge is in motion it produces an additional force which is velocity dependent. This force is called the magnetic force.

Magnetism arise from two types of motions of electrons in atoms.

1. Motion of electrons in an orbit around the nucleus.
2. Spin of electrons around its axis.



- In many elements the magnetic moments of electrons cancel each other.
- However (in transition metals)

e.g. iron (Fe), cobalt(Co), nickel(Ni) - these elements are common examples for magnetic materials
magnetic moments are not cancelled each other (net magnetization is present)

Magnetic Poles

- Magnetic poles – from which magnetic forces emanate.
- Two types: north and south

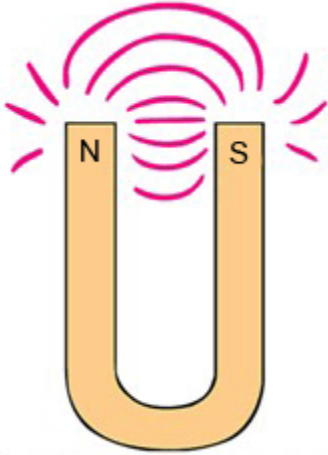
Eg: Bar magnet



- Every magnet has *both* a north and a south pole
- Like poles repel each other; unlike poles attract.
- Simple compass:

Suspend bar magnet from its center by a piece of string. Convention is that the north pole points northward; south pole points southward.

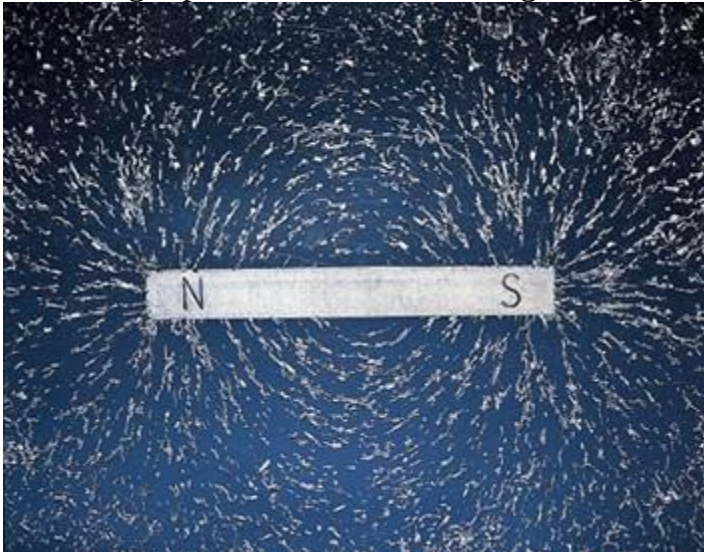
- This means that the Earth's “north pole” is actually a magnetic south pole! And vice-versa. (See the figure given in Earth's magnetic field)
- Another type: Horseshoe magnet



- Fridge magnets – have narrow alternating N and S strips. Strong field near the magnet but field decays quickly with distance since N and S fields cancel.
- Magnetic poles cannot be isolated. (Big difference with electric charge) eg if break bar magnet in two, each half behaves as complete magnet, each with N and S poles. Even when it's one atom thick! No magnetic monopoles

Magnetic Fields

Iron filings sprinkled around bar magnet align with the magnetic field



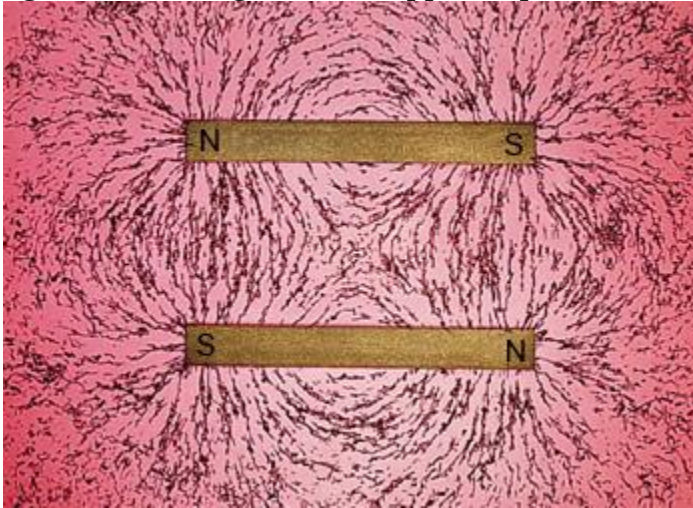
(compare with electric field lines in a earlier lesson)

Field is stronger where lines are more dense

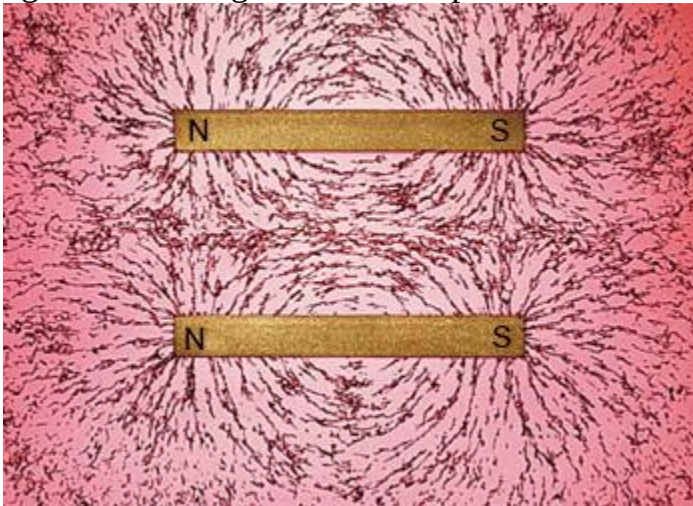
Actually magnetic field lines form closed loops – they continue inside the magnet (not shown in

pic)

Eg. Two bar magnets with opposite poles at same end



Eg. Two bar magnets with like poles at same end.



Where is the motion that makes a stationary bar magnet magnetic?

- The moving charges are the electrons – undergoing two kinds of constant motion:
 - (i) spin, like "tops" (although, really need quantum mechanics to describe this). Usually spin is the significant contribution
 - (ii) orbit (revolve) about nucleus
- Every spinning electron is a tiny magnet. Net magnetism comes from sum of fields from every electron.
- Non-magnetic materials: consists of pairs of electrons spinning in *opposite* directions, so their fields cancel each other, and there is no net magnetic field.

- Magnetic materials: eg iron, nickel, cobalt, not all spins are cancelled out. Eg. each iron atom has 4 electrons whose spin magnetism isn't cancelled

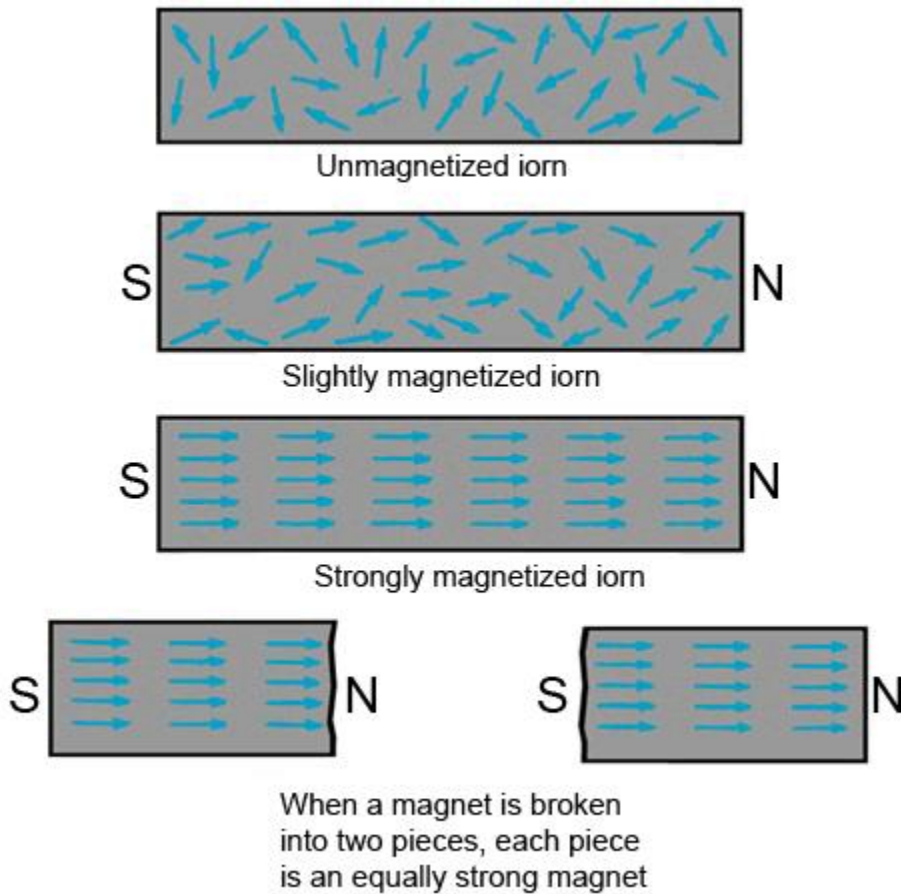
Magnetic Domains and Making Magnets

- Magnetic field of an individual iron atom is so strong, that it makes neighboring atoms line up - get clusters of billions of aligned atoms, called **magnetic domains**.



- Domains themselves can align with each other.
- A common piece of iron is not a magnet since domains are generally independent with random orientations.
- Only when they align with each other, does the piece of iron become a magnet.
- Can make them align by bringing a (permanent) magnet nearby – if this magnet is strong enough, then when you separate them again, the iron piece may retain the alignment, and so be magnetic.
- But if not strong enough, then on removing the permanent magnet, the domains in the iron piece thermally move back to a random arrangement.
- Another way to make magnet: stroke iron piece with magnet – aligns domains

Here are the stages in magnetizing a piece of iron:



Question

If a magnet is dropped or heated, does the magnetism get weaker, and why?

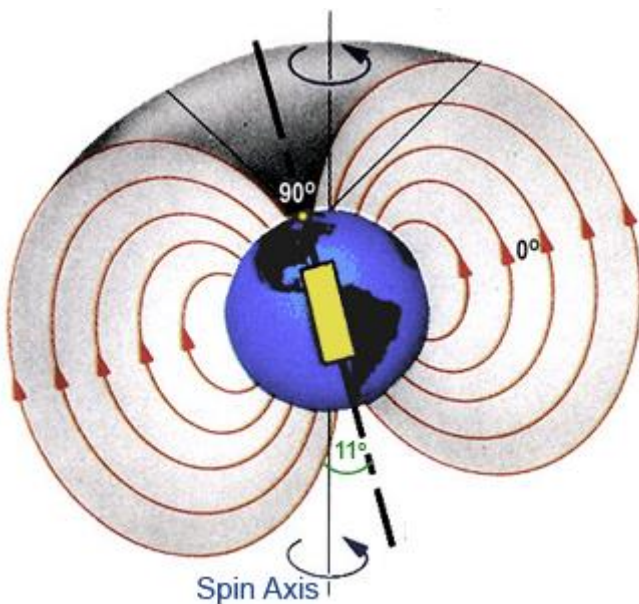
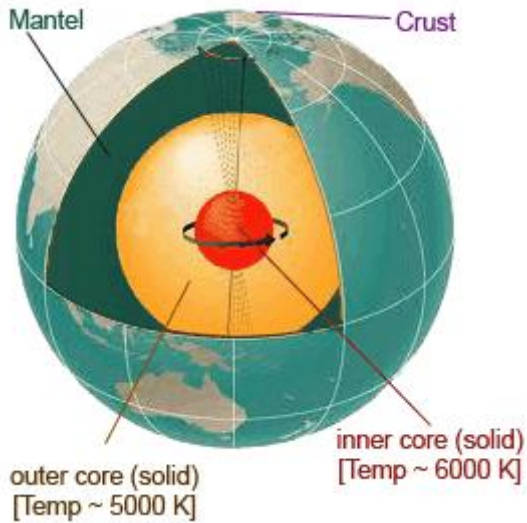
- A) No, in fact it increases since the electrons move faster.
- B) Yes, because domains get jostled out of alignment.
- C) No, it doesn't change
- D) Yes, because some of the electrons leave the magnet

Question

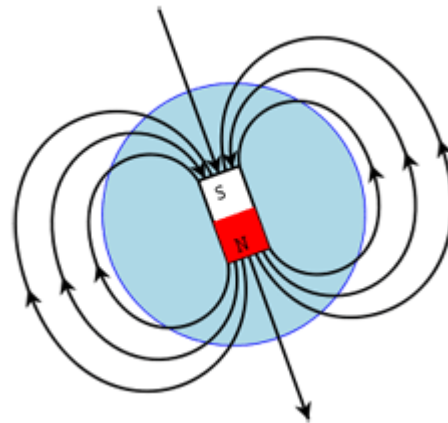
Place an unmagnetized piece of iron in a magnetic field (eg iron filings near a magnet). Why is it attracted to the magnet?

Earth's magnetic field

- The origin of earth's magnetic field is not totally understood.
- It is thought to be a result of electrical currents generated by the movement of molten iron & nickel within earth's outer core!



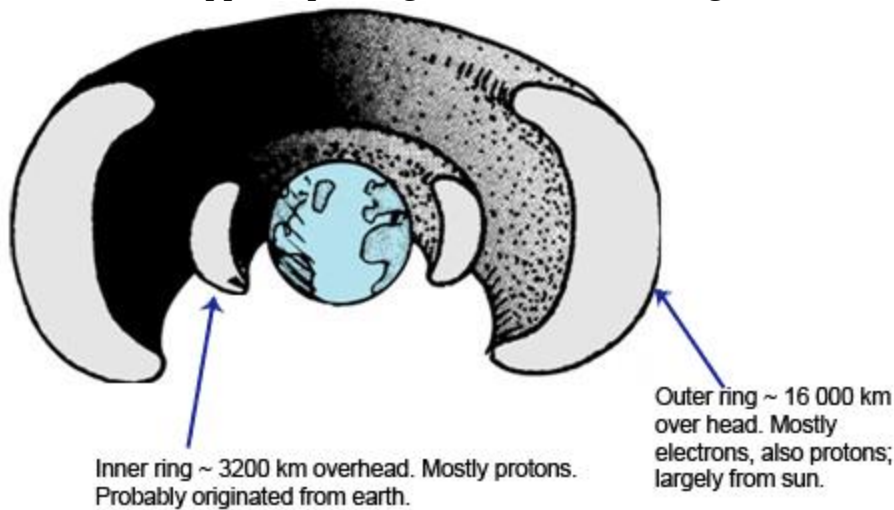
The earth's magnetic field is similar to that of a bar magnet tilted 11° from the spin axis of the earth.



- Unstable – it can diminish to zero, and then reverse direction. More than 20 such pole reversals have occurred in the past 5 million years! Next one expected within 2000 years. Can detect from magnetization within Earth's upper rock strata.....Read more!!!!
- Sun's magnetic field also reverses regularly, every 22 years. (Related to droughts on earth?)
- Smaller and faster fluctuations in Earth's field are from varying ion winds in atmosphere. Ions created from solar ultraviolet and x-rays interacting with atmospheric atoms.
- Moon has no magnetic field and therefore is thought to have a small core that is partially molten

Cosmic rays

- Are actually charged particles (protons, or other atomic nuclei), produced by sun, or other stars.
- Travel at speeds close to the speed of light
- Dangerous radiation for humans; also can mess up electronic instrumentation.
- Earth's magnetic field protects us from them by deflecting them back .
- Some are trapped, spiraling back and forth along field lines, in two "van Allen radiation belts":



- When ions dip into Earth's atmosphere, get beautiful light shows from fluorescence: *aurora borealis* (northern hemisphere), and *aurora australis* (southern hemisphere)



- Spaceflights try to avoid the belts, since hazardous to astronauts and instruments.
- Although Earth's field protect us on surface of earth from "primary" cosmic rays, we do get bombarded with "secondary" ones – when primary rays strike atomic nuclei high in atmosphere.
- Greatest bombardment at the magnetic poles, because the rays come in parallel to the field lines so don't get deflected as much as those coming in near equator, perpendicular to field.

- Mid-latitudes, about 5 particles per square cm at sea-level!

Biomagnetism

- Some animals have magnetite (iron oxide) domains in their bodies i.e. a built-in compass, allowing them to sense magnetic fields, and thus navigate!

Eg. Some bacteria, pigeons, wasps, monarch butterflies, sea turtles, bees...

- Read more on this.

Question



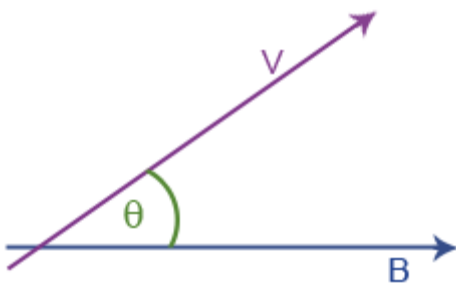
Compared to the huge force that attracts iron track to a strong magnet, the force that the track exerts on the magnet is

1. Relatively small
2. Equally huge

Motion of charge particles

Experiments on different charges of various velocities

- (1) The force is proportional to the magnitude of charge q and velocity v .
- (2) If v is parallel or anti parallel (opposite to B) to B , the force is zero.
- (3) If v is not parallel to B then there is a force perpendicular to both v and B .
- (4) If v makes an angle with B , then the force is proportional to $\sin\theta$.



(F_m – magnetic force)

$$F_m \propto qV \sin \theta$$

$$F_m = BqV \sin \theta$$

$$F_m = qV \Delta B$$

v - Velocity of charge

q - charge.

B - Magnetic flux density (magnetic flux passing through unit area of a magnetic field in a direction at right angle to the magnetic field)

Units

$$B = \frac{F}{qV \sin \theta} \left(\frac{N}{C \text{ ms}^{-1}} \right)$$

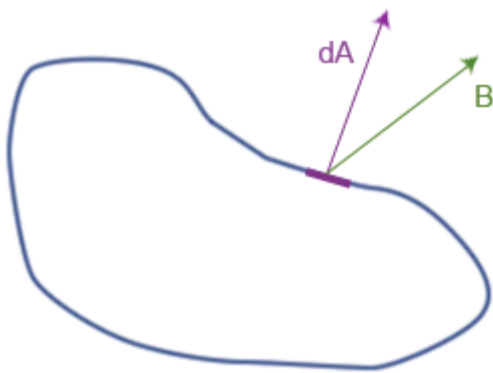
$$\left(\frac{N}{Am} \right)$$

$$(NA^{-1}m^{-1}) \rightarrow \text{Tesla} \rightarrow$$

$$10^4 \text{ Gauss} = 1 \text{ Tesh}$$

$$\text{Earths magnetic field} = 5 \times 10^{-5} T$$

Magnetic field



$$\Phi = \oint \underline{B} \cdot d\underline{A}$$

$$\Phi \rightarrow (NA^{-1} m^{-1})(m^2) \rightarrow NA^{-1}m$$

$$\rightarrow Wb (\text{weber})$$

$$\uparrow \text{Units of } \Phi$$

Motion of charged particles in a magnetic field

$$\underline{V} // \underline{B}$$

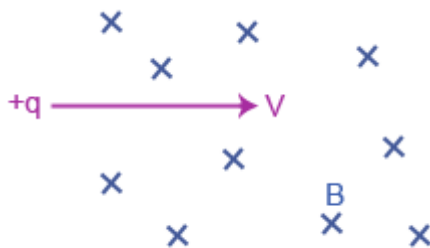


$$F = qv\Delta B$$

$$\rightarrow F = 0 \rightarrow \theta = 0, \sin \theta = 0$$

$\rightarrow$ no change in the direction (moves in a straight line)

V is Perpendicular to B



$$F = qv \Delta B$$

$$F = qv \sin 90^\circ B$$

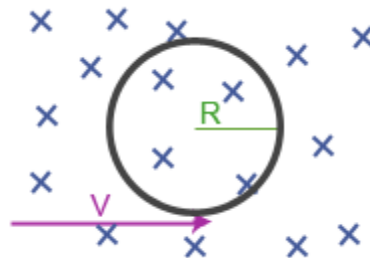
$$F = qvB$$

due to this force ($F = q V B$) the particle moves in a circular path.

$$qvB = \frac{mV^2}{R}$$

$$R = \frac{mV}{qB}$$

m – mass of the particle



V is at an angle θ with B



$$v \sin \theta \text{ (perpendicular to B) } - (1)$$

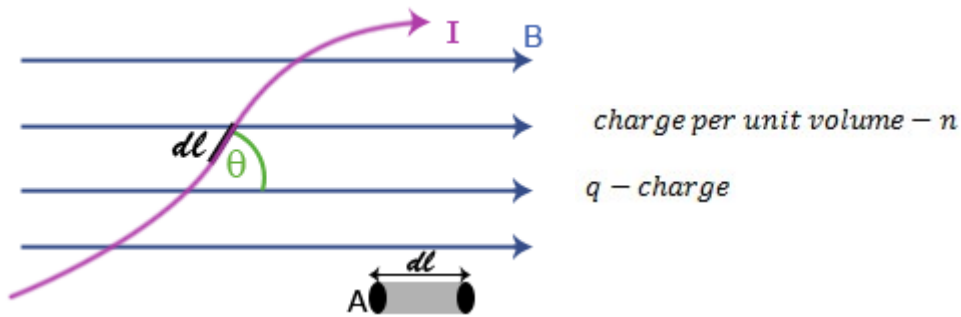
$$v \cos \theta \text{ (// to B) } - (2)$$

(1) $\rightarrow$ circular motion

(2) $\rightarrow$ linear motion

(1) & (2) $\Rightarrow$

Force on a conductor placed in an uniform field



$$\text{Charge in } (A \times dl) = (nAdl)q$$

$$[\text{force on charge } q = q(\underline{V} \Delta \underline{B})]$$

$$\therefore \text{Force on the above element } (dF) = (n A dl)q \underline{V} \Delta \underline{B}$$

$$dF = n A q dl VB \sin \theta$$

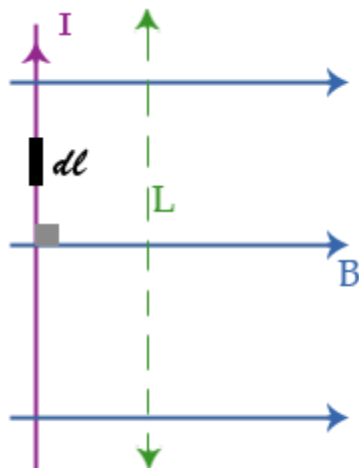
$$= B(nqV)Adl \sin \theta$$

$$= B \underbrace{J A}_{I} dl \sin \theta$$

$$dF = B I dl \sin \theta$$

$$= I(\underline{dl} \Delta \underline{B}) \quad \left. \vphantom{\int} \right\} \text{ The direction of } dl = \text{the direction of the current}$$

Linear conductor of length L



$$\text{force on } dl = dF \text{ (say)}$$

$$= I(dl \Delta B)$$

$$= I(dl B \sin \theta)$$

$$F = \int dF$$

$$F = \int I dl B \sin \theta$$

$$F = IB \sin \theta \int dl$$

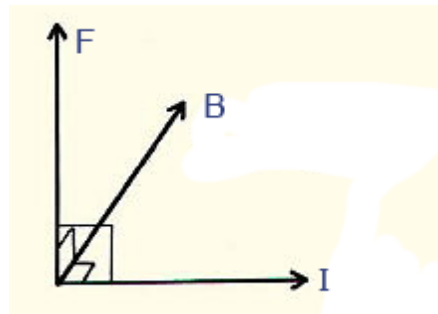
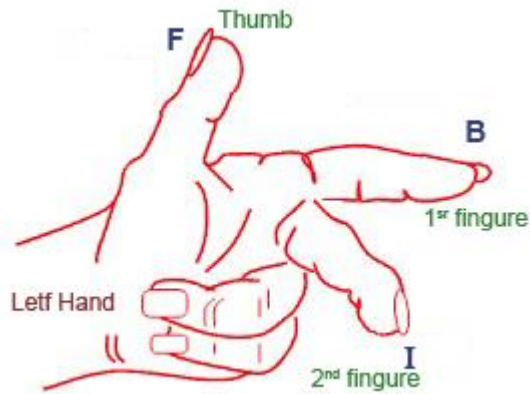
$$= IB \sin \theta L$$

When $\theta = 90$

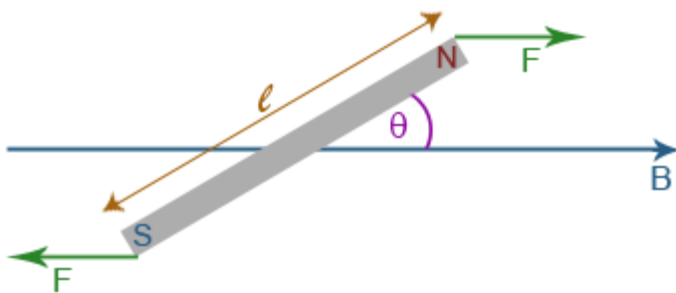
$$F = BIL$$

Fleming's Left Hand Rule

To find the deviation in which the force acts on a conductor



Torque on magnets & current loops



If a magnet is placed in a magnetic field, force acts on the poles. If B is uniform the force on each pole is same in magnitude and opposite in direction.

$$\text{Torque } \tau = F (l \sin \theta)$$

$$F = mB$$

$$\rightarrow \tau = mbl \sin \theta$$

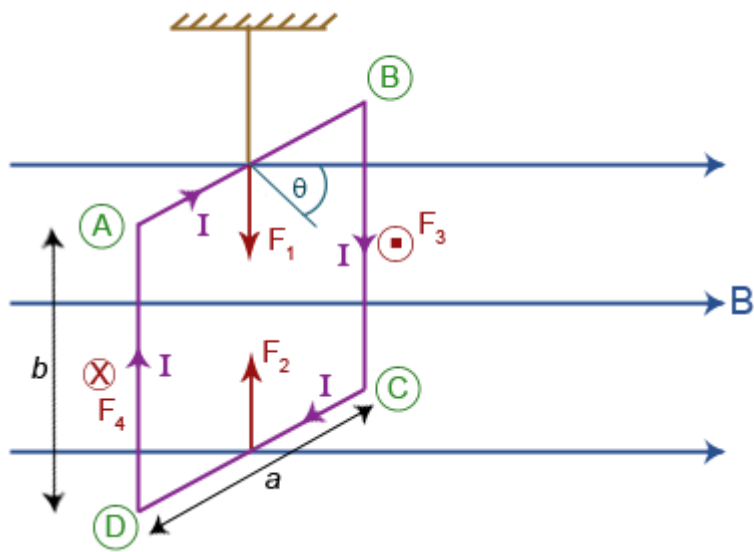
$$= MB \sin \theta$$

m – pole strength of the magnet

M = ml (magnetic moment)

(units of Magnetic moment (JT^{-1}) or (Am^2))

Current loop in a uniform B



Force on AB (F_1) = $IB \sin(90-\theta)a$

$F_1 = IaB \cos\theta$, ($\downarrow$)

Force on DC (F_2) = $IaB \cos\theta$

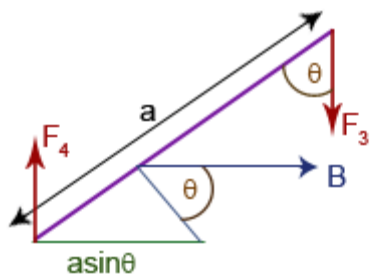
$F_2 = IaB \cos\theta$, ($\uparrow$)

$F_1 = F_2$ [cancel each other]

$F_1 = IaB \cos\theta$, ($\downarrow$)

Force on BC (F_3) = $IbB \sin 90 = IbB$

Force on AD (F_4) = $IbB \sin 90 = IbB$



$$\text{Torque on the loop} = \underbrace{(IbB)}_{\text{Force}} \underbrace{(a \sin \theta)}_{\text{distance}}$$

Force distance

$$\tau = IAB \sin \theta \quad (A = ab \text{--- area of the loop})$$

If there are N turns in the loop

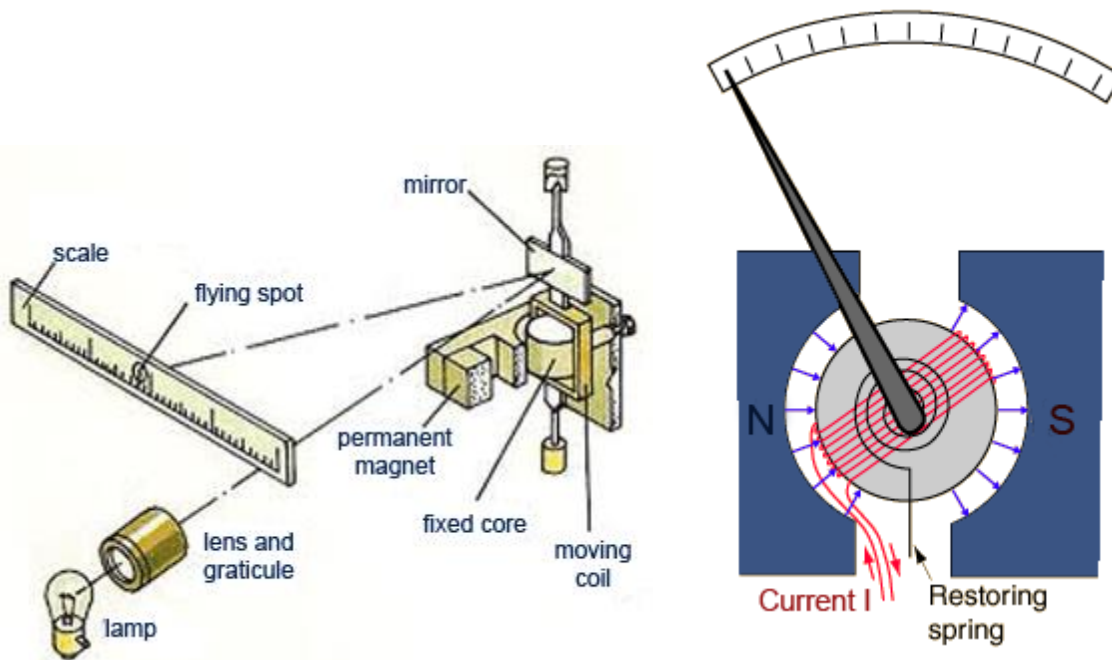
$$\tau = B \underbrace{INA} \sin \theta$$

Magnetic moment (M) $\leftarrow \text{unit } INA \rightarrow Am^2$

When $\theta = 90 \rightarrow \tau \text{ maximum}$

$\theta = 0 \rightarrow \tau \text{ minimum (0)}$

Moving coil Galvanometer



[Since there is a radial magnetic field, the plane of the coil is always parallel to B]

- When the coil carries a current the magnet exerts a torque on the coil ($\tau_1 = NIAB$; $\theta = 90$) [N – number turns]
- The coil is mounted on a spring that exerts an opposite torque (τ_2) on the coil. When both of these torques are equal the coil stops.

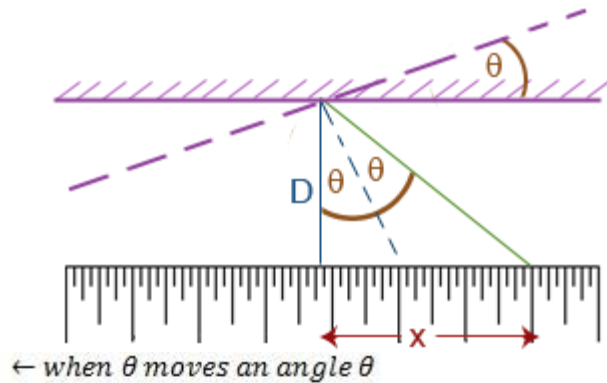
$$\tau_1 = C\theta - (1) \quad (C - \text{torsion constant of the spring})$$

$$\tau_1 = NIAB - (2)$$

$$(1) = (2) \rightarrow C\theta = NIAB \rightarrow I = \left(\frac{C}{BNA} \right) \theta - (A)$$

$$I \propto \theta$$

[θ Can be measured using the optical lever !]



$$2\theta = \tan \theta = \frac{x}{D} \quad (\because \theta \ll \ll)$$

$$\theta = \frac{x}{2D}$$

$$\rightarrow I = \left(\frac{C}{2BNAD} \right) x$$

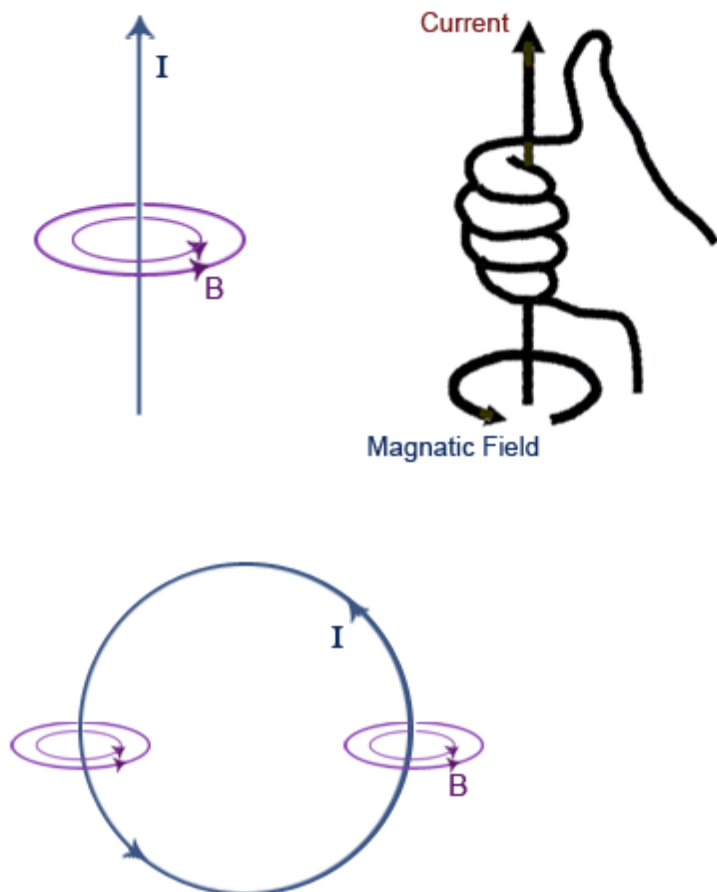
In general $D = 1m$

$$\rightarrow I \propto x$$

$$\text{Current sensitivity } K_i = \frac{x}{I} \{ \text{deflection per unit current} \}$$

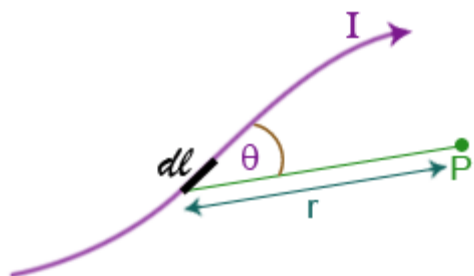
Magnetic field produced by a current carrying conductor

To find the direction of the magnetic field, right hand rule could be used.



Biot – Savart Law

- Biot Savart Law is used to determine the magnetic field at a given point due to a wire carrying a constant current.



$I dl$ – current element

Magnetic field at P due to current element

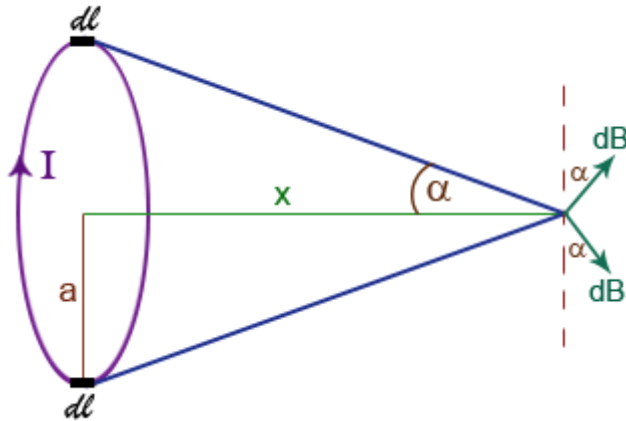
$$I dl = dB \text{ (say)}$$

$$|dB| = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$$

$$\underline{dB} = \frac{\mu_0}{4\pi} \frac{I dl \hat{r}}{r^2}$$

{ $\hat{r}$ - unit vector along r}

(1) Magnetic field on the axis of a current carrying circular loop



$$dB = \frac{\mu_0}{4\pi} \frac{I dl \hat{r}}{r^2}$$

$$= \frac{\mu_0}{4\pi} \frac{I dl \sin 90}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl}{r^2}$$

- Components perpendicular to axis cancel each other
- Components parallel to axis add!

$$\rightarrow \text{due to two opposite element (dl)} \rightarrow dB = \frac{2\mu_0}{4\pi} \frac{I dl}{r^2} \sin \alpha$$

$\therefore$ the total field due to whole ring

$$B_T = \int dB_T = \int \frac{2\mu_0}{4\pi} \frac{I dl}{r^2} \sin \alpha$$

$$= \frac{\mu_0 \sin \alpha}{4\pi r^2} \int 2dl$$

$$\begin{aligned}
 &= \frac{\mu_0 \sin \alpha I}{4\pi r^2} (2\pi a) \\
 &= \frac{\mu_0}{4\pi} \frac{a}{\sqrt{x^2 + a^2}} \frac{(2\pi I a)}{r^2} \\
 &= \frac{\mu_0}{4\pi} \frac{2\pi a^2 I}{\sqrt{x^2 + a^2} (a^2 + x^2)} \\
 B &= \frac{\mu_0}{4\pi} \frac{2\pi a^2 I}{(x^2 + a^2)^{3/2}}
 \end{aligned}$$

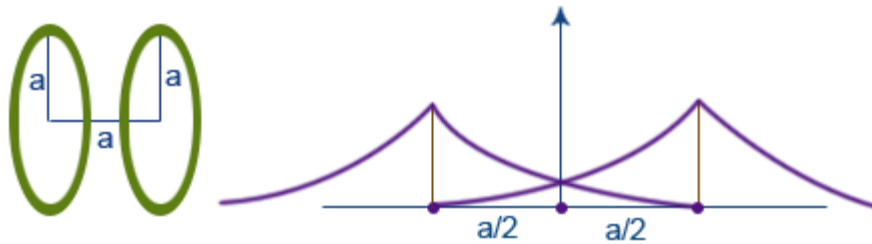
If there are N turns $B = \frac{\mu_0}{4\pi} \frac{2\pi a^2 I N}{(x^2 + a^2)^{3/2}}$ at the centre of the ring

i.e. $x=0$

$$\rightarrow B = \frac{\mu_0}{4\pi} \frac{2\pi a^2 I N}{a^3} \rightarrow B = \frac{N\mu_0 I}{2a}$$

Helmholtz coil (read!)

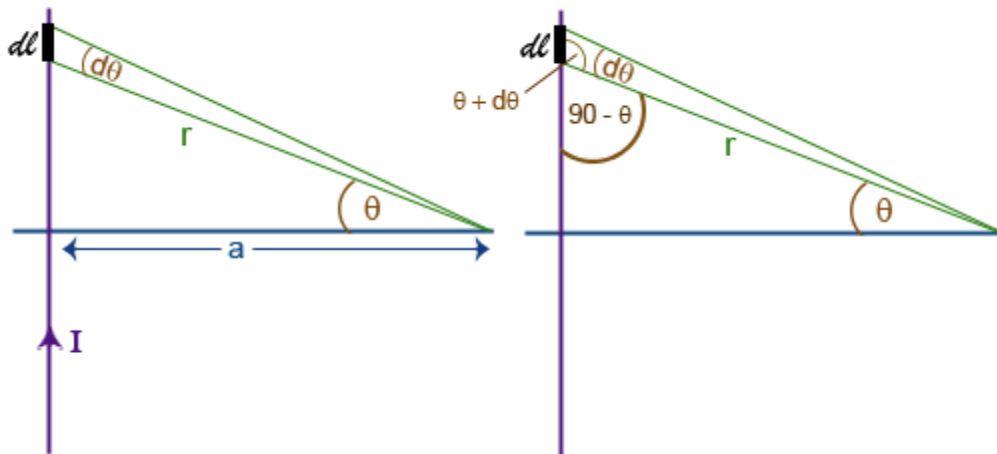
- A pair of identical cylindrical coils of wire mounted coaxially and separated by a distance equal



to the radius of the coils.

(then a uniform in magnetic field created over a considerable volume between the coils)

(2) Straight conductor



From Biot Savart law

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin (90 + \theta)}{r^2}$$

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \cos \theta}{r^2}$$

But $\cos \theta = \frac{a}{r}$, $r d\theta = dl \cos \theta$

$$\therefore dB = \frac{\mu_0}{4\pi} \frac{(I r d\theta)}{r^2} = \frac{\mu_0}{4\pi} \frac{I d\theta}{r}$$

but $r = \frac{a}{\cos \theta}$

$$\rightarrow dB = \frac{\mu_0}{4\pi} \frac{I \cos \theta d\theta}{a}$$

$$\text{Total } B = \int dB = \frac{\mu_0}{4\pi} \frac{I}{a} \int_{-\theta_1}^{\theta_2} \cos \theta \cdot d\theta$$

$$= \frac{\mu_0}{4\pi} \frac{I}{a} [\sin \theta]_{-\theta_1}^{\theta_2}$$

$$= \frac{\mu_0}{4\pi} \frac{I}{a} [\sin \theta_2 + \sin \theta_1]$$

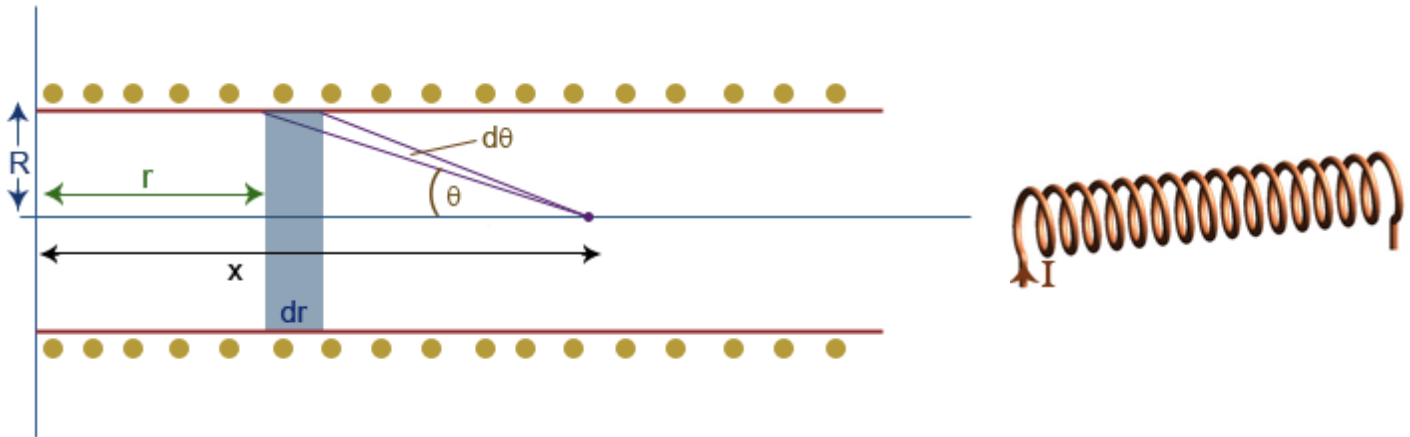
For a infinitely long conductor

$$\theta_1 = \theta_2 = \pi/2$$

$$\rightarrow B = \frac{\mu_0}{4\pi} \frac{I}{a} [2]$$

$$\rightarrow B = \frac{\mu_0 I}{2\pi a}$$

(3) Solenoid



dB – magnetic field at P due to dr

$$\rightarrow B = \frac{\mu_0}{4\pi} \frac{2\pi(n dr) R^2 I}{[R^2 + (x-r)^2]^{3/2}} \quad (A)$$

$$\text{but } x - r = \frac{R}{\tan \theta} = \frac{R \cos \theta}{\sin \theta}$$

$$\rightarrow dr = \frac{R d\theta}{\sin^2 \theta}$$

$$\rightarrow dr = R \left[\frac{-\sin \theta \sin \theta d\theta - \cos \theta \cos \theta}{\sin^2 \theta} \right]$$

$$= -R \left[\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta} \right] d\theta$$

$$= -\frac{R d\theta}{\sin^2 \theta}$$

Also $\sin \theta = \frac{R}{\left[R^2 + (z-r)^2 \right]^{1/2}}$

$$\rightarrow dB = \frac{\mu_0}{4\pi} \frac{2\pi n R d\theta}{\sin^2 \theta} \cdot \frac{R^2 I}{R^3 / \sin^3 \theta}$$

$$dB = \frac{\mu_0}{4\pi} 2\pi n d\theta I \sin \theta$$

$$dB = \frac{\mu_0 n I \sin \theta d\theta}{2}$$

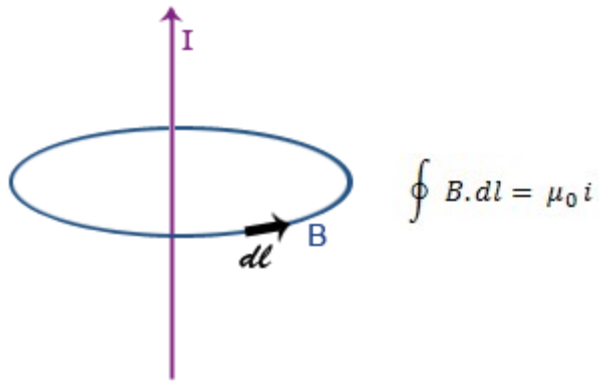
Due to the whole Solenoid

$$\begin{aligned} \rightarrow B &= \int dB = \int \frac{\mu_0 n I \sin \theta d\theta}{2} \\ &= \frac{\mu_0 n I}{2} \int_{\theta_1}^{\theta_2} \sin \theta d\theta \\ &= \frac{\mu_0 n I}{2} [-\cos \theta]_{\theta_1}^{\theta_2} \\ &= \frac{\mu_0 n I}{2} (\cos \theta_1 - \cos \theta_2) \end{aligned}$$

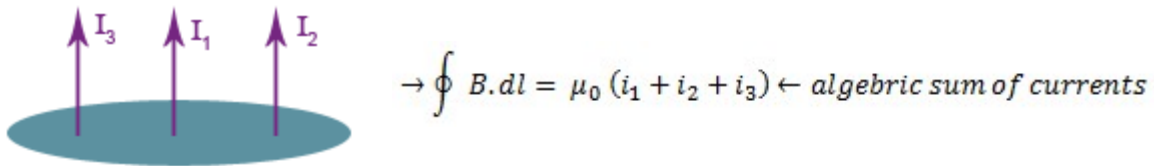
If the solenoid is infinitely long

$$\begin{aligned} \theta_1 &= 0 \text{ \& } \theta_2 = \pi \\ \rightarrow B &= \frac{\mu_0 n I}{2} [1 - [-1]] = \frac{\mu_0 n I (2)}{2} \\ B &= \mu_0 n I \end{aligned}$$

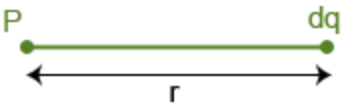
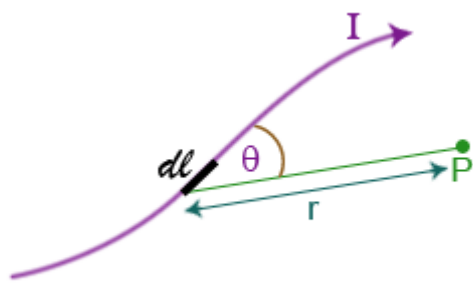
Ampere's circuital theory (Ampere's law)

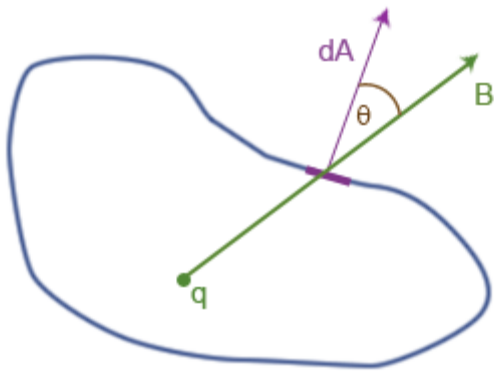
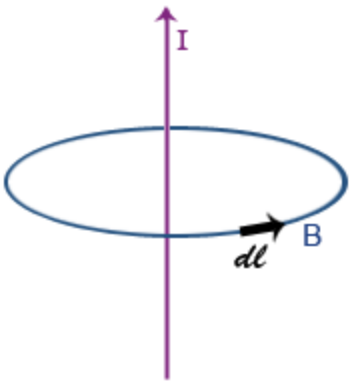


The line integral of magnetic field of an closed loop = $\mu_0 i$ (current enclosed by the loop)



Important Note

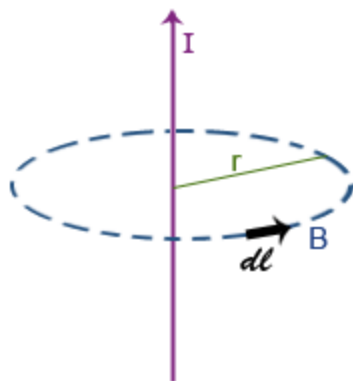
	Electricity	Magnetism
Differential form	Coulomb's law  $dE = \frac{1}{4\pi\epsilon_0} \frac{dq\hat{r}}{r^2}$	Biot - Savart law  $dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2}$

Integral form	Gauss' law  $\oint E \cdot dA = \frac{q}{\epsilon_0}$	Ampere's circular  $\oint B \cdot dl = \mu_0 I$

Eg. 01

B at a distance r from a long straight conductor

From Ampere's law



From Ampere's law

$$\oint B \cdot dl = \mu_0 I$$

$$\oint B dl = \mu_0 I$$

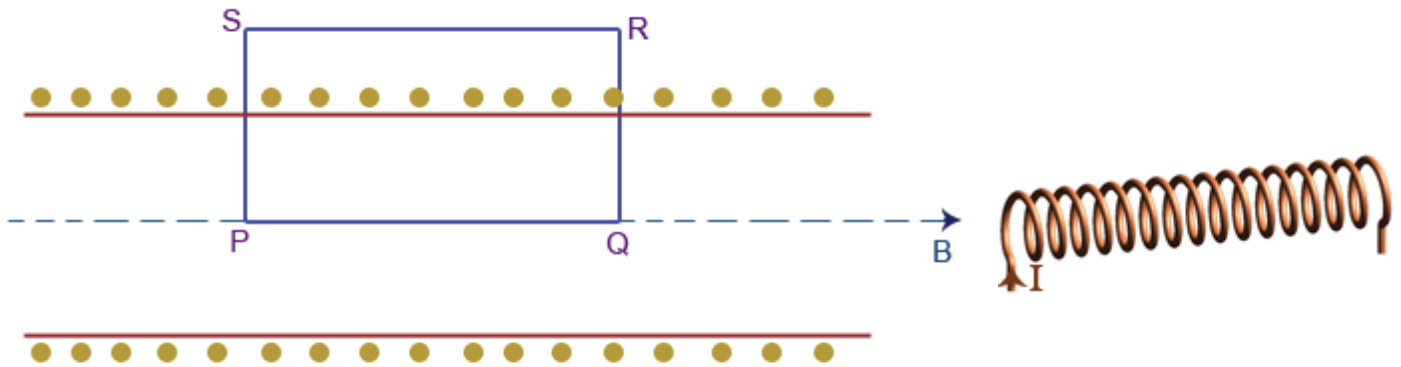
$$B \oint dl = \mu_0 I$$

$$B 2\pi r = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

Eg. 02

the centre of a Solenoid



Consider the loop PQRS;

From Ampere's law

$$\oint B \cdot dl = \mu_0 I (Nl) \text{ (line PQRS)}$$

$$\int_A^Q B \cdot dl + \int_Q^R B \cdot dl + \int_R^S B \cdot dl + \int_S^P B \cdot dl = \mu_0 INl$$

$$\int_A^Q B \cos 0 \, dl + \underbrace{\int_Q^R B \cos 90 \, dl}_{=0} + \int_R^S 0 \cdot dl + \underbrace{\int_S^P B \cos 90 \, dl}_{=0} = \mu_0 INl$$

$$= \int B \, dl = \mu_0 INl$$

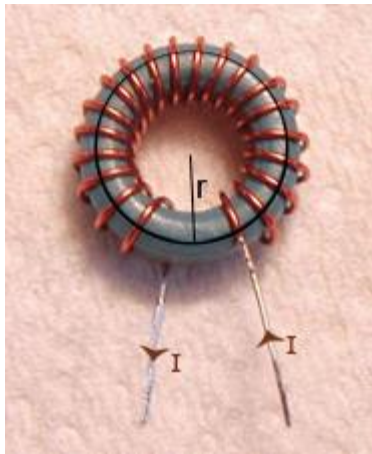
$$B \int dl = \mu_0 INl$$

$$B l = \mu_0 I N$$

$$B = \mu_0 I N$$

Eg. 03

Toroid – N – number of turns per unit length



Ampere's law $\rightarrow \oint B \cdot dl = \mu_0 I$

$$\oint B \cos \theta \, dl = \mu_0 \{2\pi r N (i)\}$$

$$\oint B \, dl = \mu_0 2\pi r N i$$

$$B \oint dl = \mu_0 2\pi r N i$$

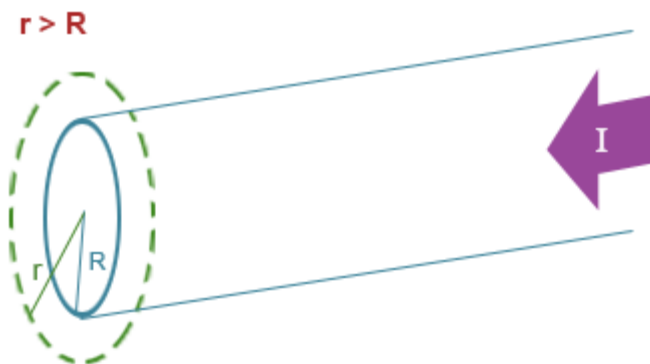
$$B 2\pi r = \mu_0 2\pi r N i$$

$$B = \mu_0 N i$$

Eg. 04

Long straight cylindrical wire of radius R.

Case I



From Ampere's law $\oint B \cdot dl = \mu_0 I$

$$\oint B \cos \theta \, dl = \mu_0 I$$

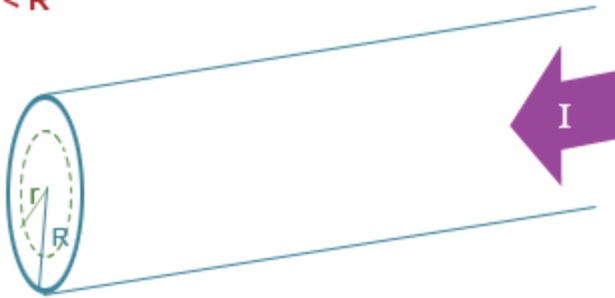
$$\oint B \cdot dl = \mu_0 I$$

$$B \oint dl = \mu_0 I$$

$$B 2\pi r = \mu_0 I \quad \rightarrow \quad B = \frac{\mu_0 I}{2\pi r}$$

Case II

$$r < R$$



Consider this loop from Ampere's law

$$\oint B \cdot dl = \mu_0 I$$

$$\oint B \cos 0 \, dl = \mu_0 \left(\frac{I r^2}{R^2} \right)$$

$$\oint B \, dl = \mu_0 \frac{I r^2}{R^2}$$

$$B = \mu_0 \frac{I r^2}{R^2}$$

$$B 2\pi r = \mu_0 \frac{I r^2}{R^2}$$

$$B = \frac{\mu_0 I r}{2\pi R^2}$$

$$\pi R^2 \rightarrow I$$

$$\pi r^2 \rightarrow ? (I^1 \text{ say})$$

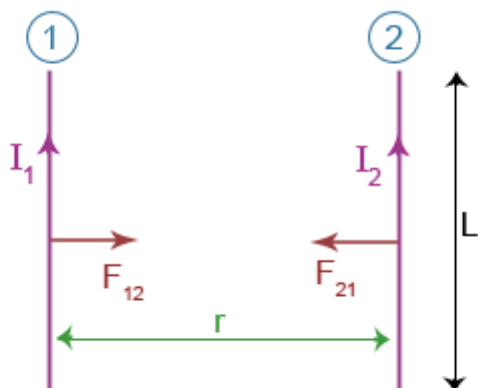
$$\frac{I \times \pi r^2}{\pi R^2} = I^1$$

$$\frac{I r^2}{R^2} = I^1$$

Force on current carrying conductors

Eg. 01

Two parallel straight conductors of length



Magnetic field on (2)

$$\text{By (1)} \quad B_{21} = \frac{\mu_0 I_1}{2\pi r}$$

Force on (2) by (1) $F_{21} = B_{21} I_2 L$

$$F_{21} = \frac{\mu_0 I_1 I_2 L}{2\pi r}$$

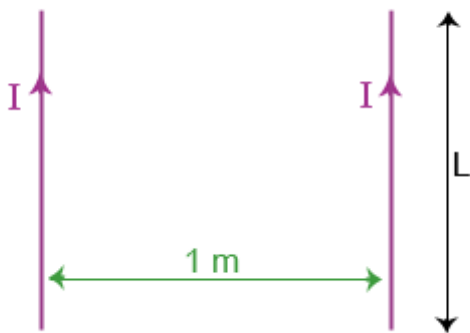
$$F_{21} = \frac{\mu_0 I_1 I_2 L}{2\pi r}$$

Similarly

- If I_1 & I_2 are in the same direction force is attractive !
- If I_1 & I_2 are in opposite directions force is repulsive !

Problem

Two current carrying straight conductors are placed in vacuum 1 m apart (see figure)



If the force on each conductor per unit length is $2 \times 10^{-7} \text{ Nm}^{-1}$, what is I ? (in Amperes)

$$\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$$

$$F_{12} = F_{21} = \frac{\mu_0 \times I \times I \times L}{2\pi r}$$

$$\frac{F}{L} = \frac{\mu_0 I^2}{2\pi r}$$

$$2 \times 10^{-7} \text{ Nm}^{-1} = \frac{\mu_0 I^2}{2\pi r}$$

$$\rightarrow I^2 = \frac{4\pi \times 10^{-7} \text{ Nm}^{-1} (r)}{\mu_0}$$

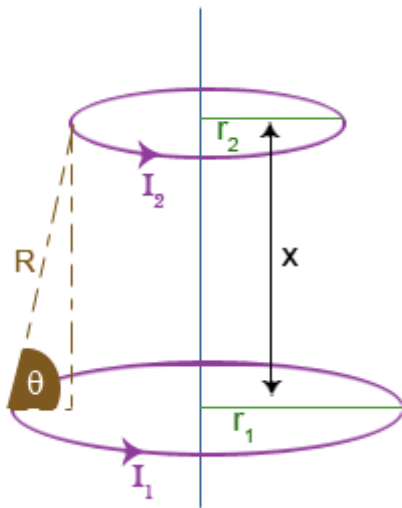
$$= \frac{4\pi \times 10^{-7} \text{ Nm}^{-1} (1\text{m})}{4\pi \times 10^{-7} \text{ NA}^{-2}}$$

$$I^2 = 1 \text{ A}$$

$$\rightarrow I = 1 \text{ A}$$

Eg. 02

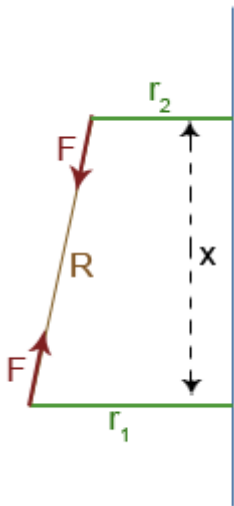
Force between two //^l circular coils of almost same radius. (coaxial coils)



$$r_1 > r_2 \text{ but } r_1 \simeq r_2$$

Force per unit length

$$f = \frac{\mu_0}{4\pi} \left(\frac{2I_1 I_2}{R} \right)$$



- All the components parallel to the plane of the coil cancel each other (by symmetry). Resultant force is parallel to the dissection of the axis
- That force

$$F = f \sin \theta (2\pi r)$$

$$r_1 \simeq r \text{ (say)}$$

$$f = \frac{\mu_0}{4\pi} \left(\frac{2I_1 I_2}{R} \right) \frac{x}{R} (2\pi r)$$

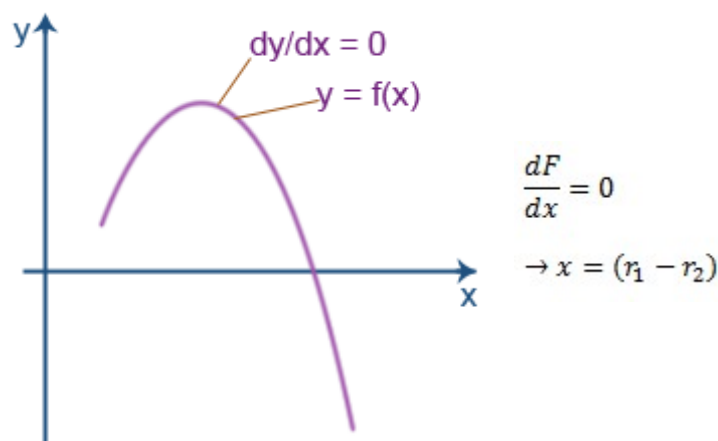
$$f = \frac{\mu_0 I_1 I_2}{4\pi} \frac{2x}{R^2} 2\pi r$$

$$\text{But } R^2 = x^2 + (r_1 - r_2)^2$$

$$\rightarrow f = \frac{\mu_0 I_1 I_2}{4\pi} \frac{2x \cdot 2\pi r}{[x^2 + (r_1 - r_2)^2]}$$

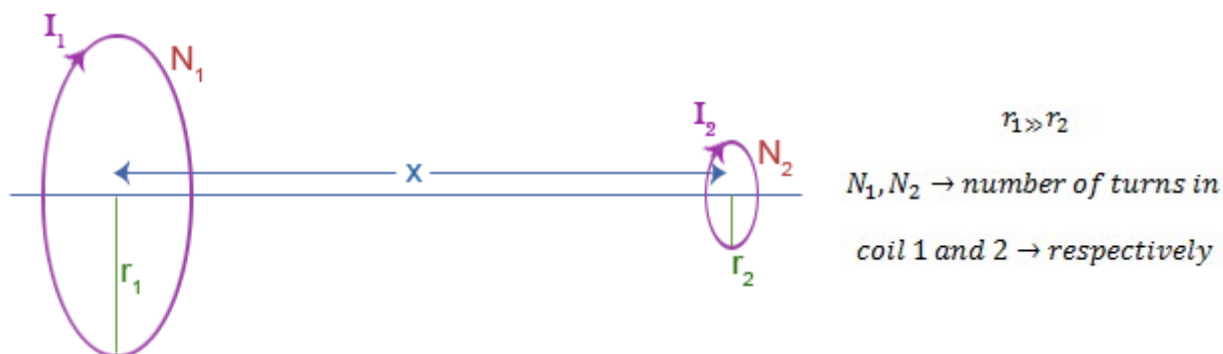
$$\rightarrow f = \frac{\mu_0 I_1 I_2 x r}{x^2 + (r_1 - r_2)^2}$$

To find the x where you get the maximum force.



Eg. 03

Two coaxial coils one having the radius very small compared to the other.



Magnetic field at a distance x from the coil

$$B_x = \frac{\mu_0 I_1 r_1^2 N_1}{2(x^2 + r_1^2)^{3/2}}$$

$$F = \frac{3\mu_0 N_1 N_2 I_1 I_2 r_1^2 r_2^2 x}{2(x^2 + r_1^2)^{5/2}}$$

$$F \text{ is maximum when } x = \frac{r_2}{2}$$

Question

Ring around what ???



A magnetic field line circles around and joins its own tail, thus forming a ring. What would you expect to find passing through some place in the region enclosed by the ring?

- (A) An electric field line
- (B) An electric current
- (C) A charging electric field line
- (D) An electric current and/or a charging electric field line

Think !!!

Is it credible that there could be a universe in which there were electric fields, but no charged bodies?

- (A) Yes, such a universe is credible
- (B) No, such a universe could never be

When a magnetic field increases or decreases – electric field is created

When an electric field increases or decreases – magnetic field is created

Collapsing or drying fields of one kind induce new fields of the other kind.

So on and on its goes – one handing off to the other

The repeated hand – off is called an electromagnetic wave

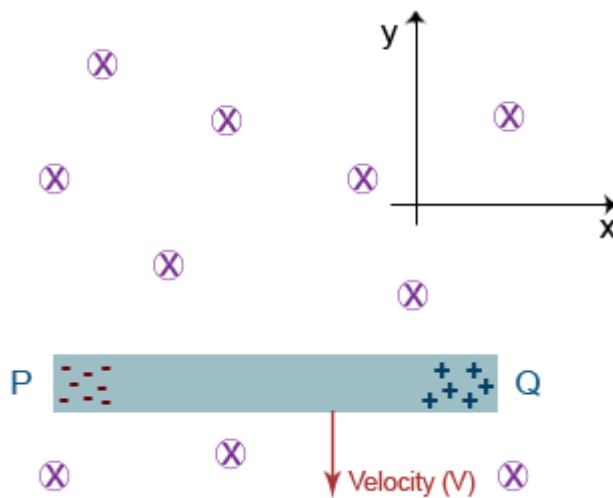
One it gets started it is self- sustaining and in no way depends on the original charge that the made the first electric field that got it all started.

A radio transmitter or star may be destroyed, but once the electric and magnetic fields in the radio signal or starlight are set on their way, they will go for millions of years regardless of the fate of their home source.

In particular, you cannot have charged bodies without also having fields. But you can have fields without charged bodies. Apparently the fields are more basic than the charges!! How strange. People use to think the charges were the real thing and the fields abstraction – and now it is the other way around

Electromagnetic Induction

Consider a conductor (PQ) moving with a velocity v perpendicular to a magnetic field



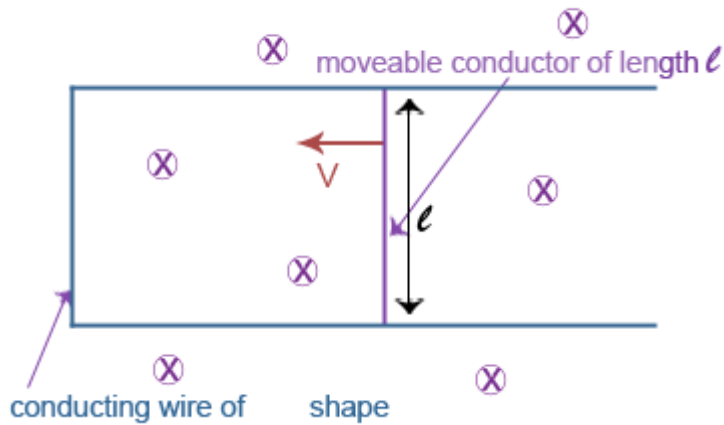
The free electrons in the conductor are moving with a velocity v in $-y$ direction. $\rightarrow$ electrons are moving towards P ($F=q\mathbf{V} \times \mathbf{B}$)

$\rightarrow$ effective negative charge at " P end"

$\rightarrow$ effective positive charge at " Q end"

$\rightarrow$ emf is induced

The magnitude and the direction of induced emf



- The area sweeping by the conductor in time $dt = (vdt)l$
- The number of field lines cutting by the movable conductor in time $dt = d\theta = Bvdtl$

$$\rightarrow \frac{d\theta}{dt} = Bvl = E \leftarrow \text{induced emf}$$

Laws of electromagnetic induction

1. Faraday's Law

The magnetic of induced emf $|E| = \left| \frac{d\theta}{dt} \right| \leftarrow \text{rate of change of magnetic flux}$

2. Lenz's Law

The direction of the induced emf is such that it opposes the change in magnetic field

$$ie E = -\frac{d\theta}{dt}$$

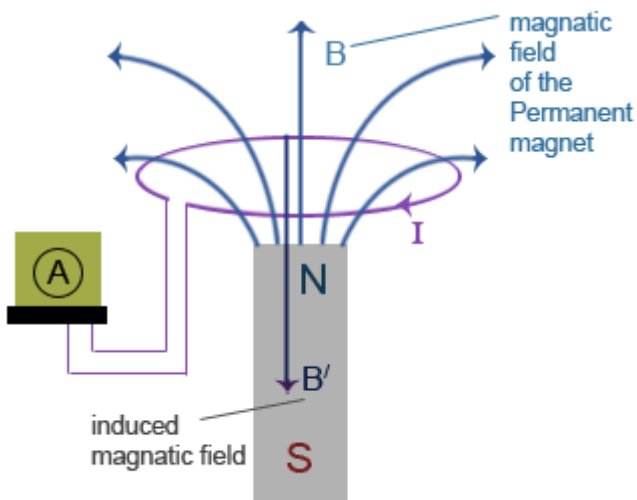
Problem



An airplane travels with a speed of 1000 kmh^{-1} in a region where the Earth's magnetic field is $5 \times 10^{-5} \text{ T}$ (consider that the Magnetic Field is nearly vertical) what is the emf induced between the wing – tips which are 70m apart

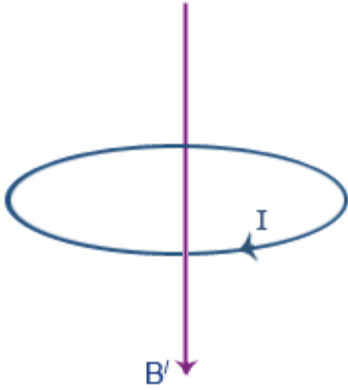
use $E = Blv$

- Further experiment (by Faraday) showed that an emf is induced in a conductor when placed in a region of varying magnetic flux.
- Consider a loop (made of conducting material) and a permanent magnet



- When the magnet is moved up, flux through the coil increases!

- Therefore according to the **Lenz's law** induced emf should oppose the change in the magnetic flux (i. e induced current is such that it reduce the magnetic flux $\rightarrow$ induced magnetic field (B') should point downwards! $\rightarrow$ the direction of the induced current can be found from the **Right Hand rule**



Question

Faraday's Paradox

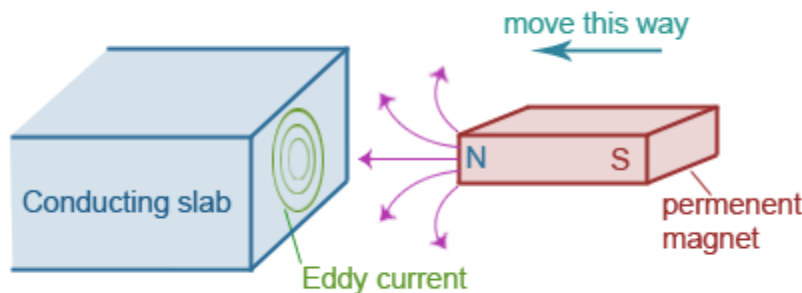
This is a coil of wire with a hunk of iron locked in it.

- (A) If current is made to flow in the wire, the iron becomes magnet.
- (B) If the iron is a magnet, current is made to flow in the wire.
- (C) Both of the first two statements are true.
- (D) Both of the two statements are false.

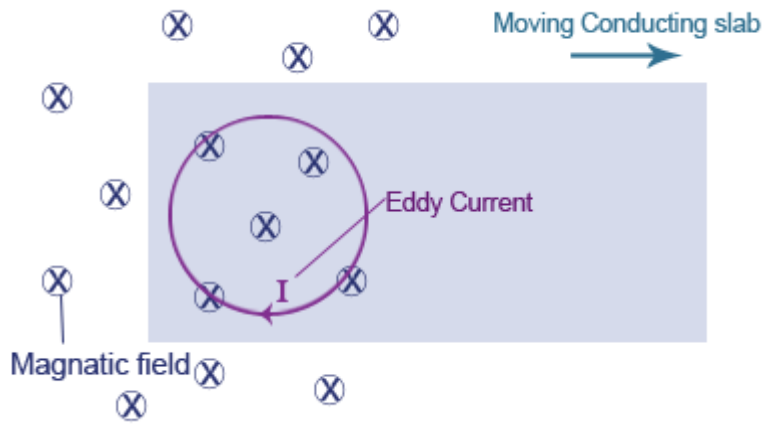
Eddy current

(there are two ways of getting induced Eddy currents)

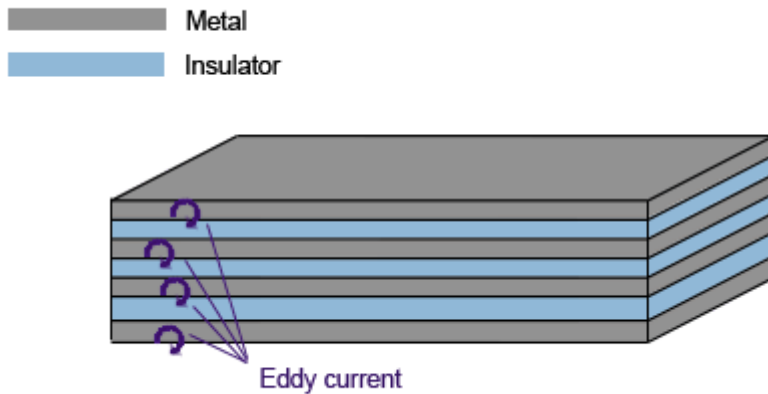
1. A current induced in a conductor when subject to a varying magnetic field.



2. A current induced in a moving conductor in a magnetic field.



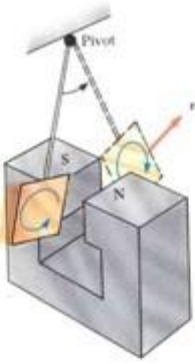
- Eddy currents are a source of energy dissipation (Eddy current loss) in alternating current (AC) Machinery.
- Via Ohmic loss – Power dissipation arising from the resistance
- To reduce the Eddy current loss the slab is made of strips of metal (pasted) together with insulating material



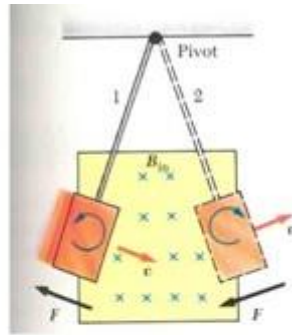
→ Eddy currents are confined to the strips!

→ large current loops are not possible thereby reducing the Eddy current loss!

Also Eddy current can be easily demonstrated by allowing a flat metal plate at the end of a rigid bar to swing as a pendulum through a magnetic field.



The formation of eddy currents in a conductor moving through a magnetic field.



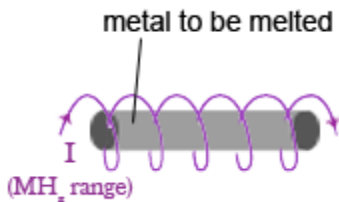
As the conducting plate enters the Field in position 1, the eddy currents are counter clockwise. In position 2, The currents are clockwise. In either case, The plate is repelled by the magnet and eventually comes to rest



When slots are cut in the Conducting plate, the eddy Currents are reduced.

Induction Heating (one of the uses of Eddy currents)

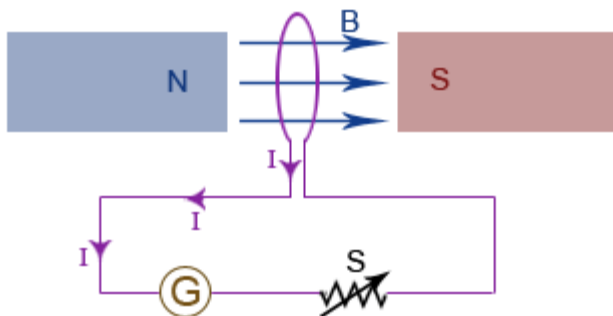
Induction heating can be used for metal melting. The advantage is that the heat is generated in the metal itself.



There are many other uses of Eddy currents!

Search coil (Exploring coil)

- A coil used for measuring magnetic flux. (magneto meter)



- It has N number of turns

- r is the total resistance of the coil
- A is the area of the coil
- G is the Galvanometer resistance
- S is the external resistance

The total flux through the coil $\Phi = BNA$

When the coil is removed from the magnetic field ; magnetic flux through the coil $\Phi_{\text{final}} = 0$

$$\text{Induced emf } E = -\frac{d\Phi}{dt} \quad (1)$$

$$\text{Induced current } i = \frac{E}{(r+s+G)} \rightarrow r + s + G = R \leftarrow \text{total resistance}$$

$$\begin{aligned} (1) \ \& \ (2) \rightarrow i &= -\frac{1}{(r+s+G)} \frac{d\Phi}{dt} \\ \rightarrow \int i \, dt &= -\frac{1}{(r+s+G)} \int_{\Phi_{\text{initial}}}^{\Phi_{\text{final}}} d\Phi \\ \rightarrow q &= -\frac{1}{(r+s+G)} [\theta_{\text{final}} - \theta_{\text{initial}}] \\ q &= -\frac{1}{(r+s+G)} [0 - BNA] \\ q &= \frac{BNA}{r+s+G} \quad (3) \end{aligned}$$

But the charge Sensitivity of the Galvanometer can be defined as

$$K_q = \frac{\theta_{\text{maximum}}}{q} \quad (4)$$

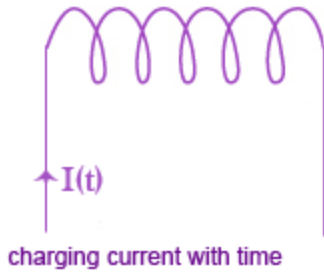
Where K_q – charge sensitivity

θ_{maximum} – Maximum deflection of the Galvanometer

q – charge

$$\begin{aligned} (2) \ \& \ (4) \rightarrow \frac{\theta_{\text{max}}}{K_q} &= \frac{BNA}{r+s+G} \\ \rightarrow B &= \frac{\theta_{\text{max}}[r+s+G]}{K_q NA} \end{aligned}$$

Self inductance & mutual inductance



Self inductance

If a changing current (eg. ac current) is flowing through a coil, it produces a changing magnetic field (B) $\rightarrow$ changing magnetic flux $\Phi \rightarrow$ opposing emf

$$\Phi \propto I$$
$$\rightarrow \Phi = LI \quad (L - \text{constant})$$

L - self coefficient of inductance of the coil

Units of L

$$\text{Henry}(H) = \frac{Wb}{A} = \frac{Tm^2}{A}$$

$$L = \frac{\Phi}{I} = \frac{Nm^{-1}}{A}$$

$$\Phi = BA$$

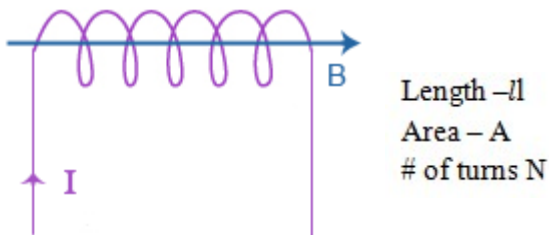
$$Wb = Tm^{-2}$$

$$\text{- induced emf } \mathcal{E} = -\frac{d\Phi}{dt} = -\frac{d(LI)}{dt} = -L \frac{dI}{dt}$$

- this induced emf opposes original emf which supplied to the coil!

- if for a coil L is significant, then the coil is called an inductor

Calculation of L for a long solenoid.



$$\Phi = NAB$$

$$\text{but } B = \mu_0 \left(\frac{N}{l} \right) i$$

$n = \# \text{ of turns for unit length}$

We've calculated this in early lessons

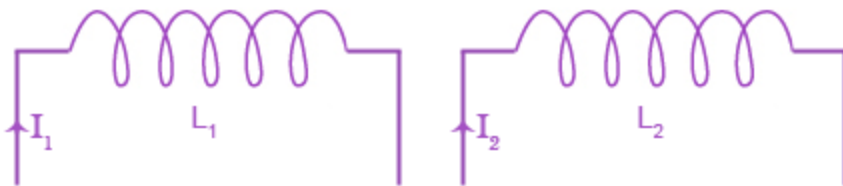
$$\rightarrow \Phi = NA\mu_0 \left(\frac{N}{l} \right) i$$

$$\text{but } L = \frac{\Phi}{i} = NA\mu_0 \frac{\left(\frac{N}{l} \right) i}{i} = \frac{N^2 A \mu_0}{l}$$

$$L = \frac{N^2 A \mu}{l} = n^2 A \mu_0 l$$

$$\left[\frac{N}{l} = n \right] \# \text{ of turns per unit length}$$

Mutual Inductance



Magnetic flux in coil (2) due to coil (1)

$$\Phi_{21} \propto i_1 \rightarrow \Phi_{21} = M i_1 \rightarrow M = \frac{\Phi_{21}}{i_1}$$

(M – unit – Henry)

Where M – coefficient of mutual inductance

Magnetic flux in coil (1) due to coil (2)

$$\Phi_{12} = M i_2$$

M – Varies only on

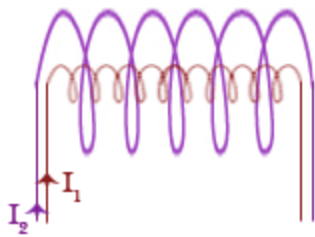
- Size of the coil
- Number of turns
- Distance between 2 coils

Total flux through coil (2)

$$\Phi_{\text{total}} = L_2 i_2 + M i_1$$

Calculation of M

(01) 2 coaxially placed solenoids of different radii



Small coil (1)

$$r_1, N_1, I_1, l, A_1$$

Large coil (2)

$$r_2, N_2, I_2, l, A_2$$

For small coil

$$B_1 = \mu_0 \frac{N_1 I_1}{l}$$

Flux on coil (2) due to coil (1) $\Phi_{21} = N_2 A_1 B_1$

$$\therefore \Phi_{21} = \frac{N_2 A_1 \mu_0 N_1 I_1}{l}$$

$$\therefore \frac{\Phi_{21}}{I_1} = \frac{N_1 N_2 A_1 \mu_0}{l} = M_{21} \text{ - (A)}$$

For large coil

$$B_2 = \frac{\mu_0 N_2 I_2}{l}$$

Flux on coil (1) due to coil (2)

$$\Phi_{12} = N_1 A_1 B_2$$

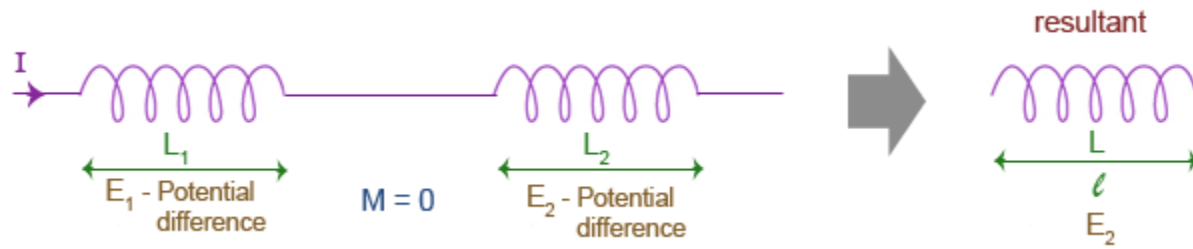
$$= \frac{N_1 A_1 \mu_0 N_2 I_2}{l}$$

$$\frac{\Phi_{12}}{I_2} = \frac{N_1 N_2 A_1 \mu_0}{l} = M_{12}$$

$$(A) \& (B) \rightarrow M_{12} = M_{21} = M$$

(02) 2 Inductors in series

Case I two inductors are far apart



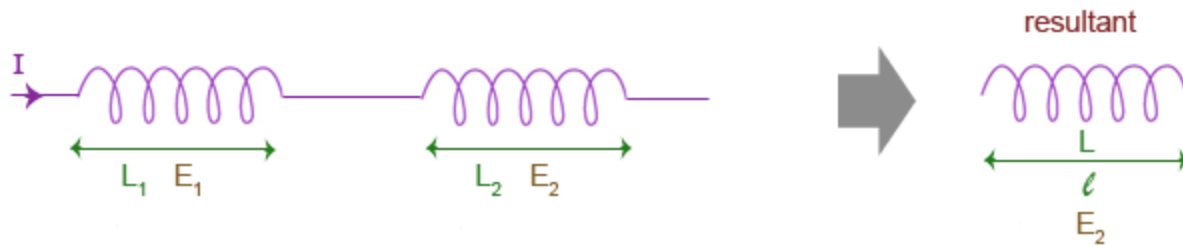
Total potential difference $E = E_1 + E_2$ — (1)

We know $E = -L \frac{di}{dt}$

(1) $\rightarrow -L \frac{di}{dt} = -L_1 \frac{di}{dt} + -L_2 \frac{di}{dt}$

$\rightarrow L = L_1 + L_2$

Case II two inductors are closely apart



$E_1 = -L_1 \frac{di}{dt} \pm M \frac{di}{dt}$

(-) comes if the coils are wound in the same direction

(+) comes if the coils are wound in opposite direction

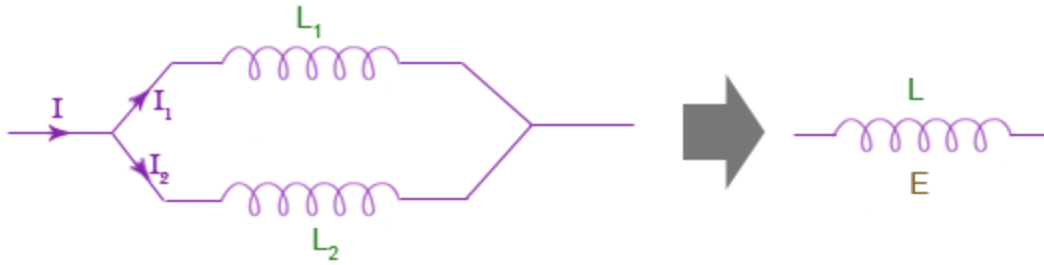
$E_2 = -L_2 \frac{di}{dt} \pm M \frac{di}{dt}$

$E = E_1 + E_2$

$-L \frac{di}{dt} = -L_1 \frac{di}{dt} \pm M \frac{di}{dt} - L_2 \frac{di}{dt} \pm M \frac{di}{dt}$

$L = L_1 + L_2 \pm 2M$

(03) 2 inductors in parallel



Case I two coils are apart $M = 0$

$$i = i_1 + i_2$$

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} \rightarrow (A)$$

❖ For the resultant

$$E = -L \frac{di}{dt} \quad - (1)$$

❖ For coil (1)

$$E = -\frac{L_1 di_1}{dt} \quad - (2)$$

❖ For coil (2)

$$E = -\frac{L_2 di_2}{dt} \quad - (3)$$

(A), (1), (2), & (3) $\rightarrow$

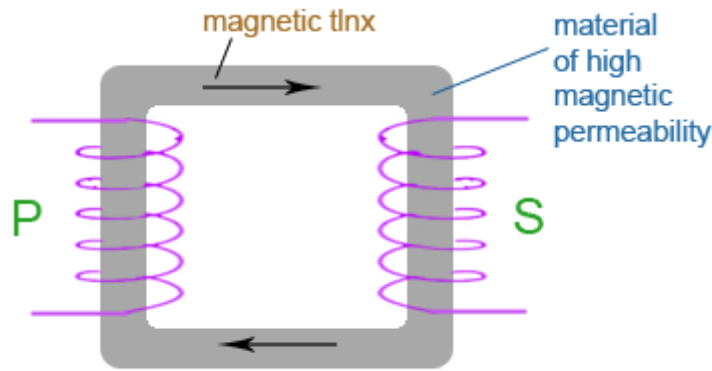
$$-\frac{E}{L} = \frac{-E}{L_1} + \frac{-E}{L_2} \rightarrow \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2}$$

Case II two coils are closely apart

$$M \neq 0$$

$$i = i_1 + i_2$$

$$\frac{di}{dt} = \frac{di_1}{dt} + \frac{di_2}{dt} \rightarrow *$$



E across coil (1)

$$E = -\frac{L_1 di_1}{dt} \pm \frac{M di_2}{dt} \quad (1)$$

E across coil (2)

$$E = -\frac{L_2 di_2}{dt} \pm \frac{M di_1}{dt} \quad (2)$$

(1) - (2)

$$0 = (L_2 \pm M) \frac{di_1}{dt} - (L_1 \pm M) \frac{di_2}{dt}$$

$$\frac{di_2}{dt} = \frac{(L_1 \pm M) di_1}{(L_2 \pm M) dt}$$

Substitute in (1)

$$E = -\frac{L_1 di_1}{dt} \pm M \left\{ \frac{L_1 \pm M}{L_2 \pm M} \right\} \frac{di_1}{dt}$$

$$E = -\frac{di_1}{dt} \left\{ \pm M \frac{(L_1 \pm M)}{(L_2 \pm M)} \right\} - L_1$$

$$E = -\frac{di_1}{dt} \left\{ \pm M \frac{(L_1 \pm M) - L_1 L_2 \neq L_1 M}{(L_2 \pm M)} \right\}$$

$$E = -\frac{di_1}{dt} \frac{(M^2 - L_1 L_2)}{(L_2 \pm M)}$$

$$\rightarrow \frac{di_1}{dt} = \frac{E(L_2 + M)}{(M^2 - L_1 L_2)} - (A)$$

$$\frac{di_1}{dt} = \frac{E(L_1 + M)}{(M^2 - L_1 L_2)} - (B)$$

If the resultant inductance = L

$$E = -\frac{L di}{dt} \rightarrow \frac{di}{dt} = \frac{E}{-L} - (C)$$

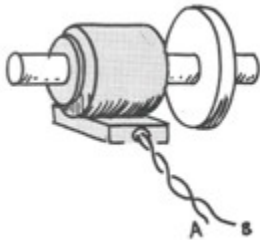
Substitute (A), (B), & (C) in (*)

$$\begin{aligned} \frac{di}{dt} &= \frac{di_1}{dt} + \frac{di_2}{dt} \\ -\frac{E}{L} &= \frac{E(L_2 + M)}{(M^2 - L_1 L_2)} + \frac{E(L_1 + M)}{(M^2 - L_1 L_2)} \\ L &= \frac{(L_1 L_2 - M^2)}{L_1 + L_2 + 2M} \end{aligned}$$

Question

A direct current motor turns clockwise when A is connected to (+) and B to the minus (-) side of a battery (the motor contains no permanent magnets).

Now if we interchange A and B so that B is plus and A is minus



- (A) The motor will turn anticlockwise
- (B) The motor will continue to clockwise

Question

Why it is hard to push a magnet down into a coil of many loops?



Question

How could a light bulb near, but not touching, an electromagnet (a current – carrying coil) be lit? Is ac or dc required?